

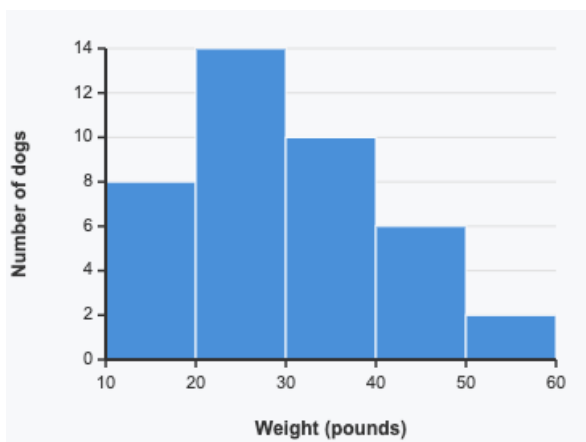
Lesson Objective

Recognize that a histogram allows us to see characteristics of the distribution that are hard to see in the raw data.

Materials: Straightedge

1. Interpret a histogram to answer questions about a data set.

The histogram shows the weights, in pounds, of dogs at a dog park. Each bar includes the left-end value but not the right-end value. For example, the first bar includes dogs that weigh 10 pounds and 15 pounds, but not 20 pounds.



- How many dogs weigh at least 30 pounds?
- How many dogs weigh exactly 25 pounds?
- What is a typical weight for a dog at this park?

Show students the histogram and point out that the x-axis shows weight in pounds and the y-axis shows the number of dogs in each interval.

Tell students that each bar includes the left-end value but not the right-end value, so a dog that weighs exactly 20 pounds is counted in the 20 to 30 pound bar, not the 10 to 20 pound bar.

Guide students through part (a). Ask which bars represent dogs that weigh at least 30 pounds, then have students add the frequencies of those bars.

For part (b), ask students to find the bar that includes 25 pounds. Ask whether they can tell from the histogram exactly how many dogs weigh 25 pounds. Point out that the histogram only tells us how many dogs fall in the 20 to 30 pound range, not the exact weights.

Answer: a) 18 dogs; b) Cannot be determined from the histogram (somewhere between 0 and 14 dogs); c) Answers vary. Sample response: about 25 pounds, since the tallest bar shows that the most dogs weigh between 20 and 30 pounds.

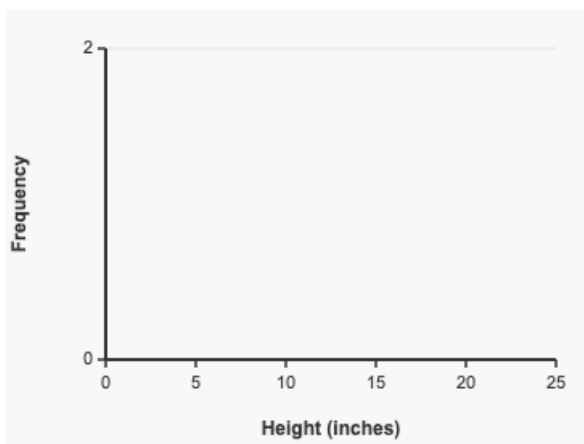
For part (c), ask students to find where most of the data falls and use that to describe a typical weight.

2. Use a frequency table to create a histogram.

The frequency table shows the heights, in inches, of plants in a garden. Use the table to create a histogram on the grid.

Height (inches) Frequency

0–5	3
5–10	7
10–15	12
15–20	5
20–25	2



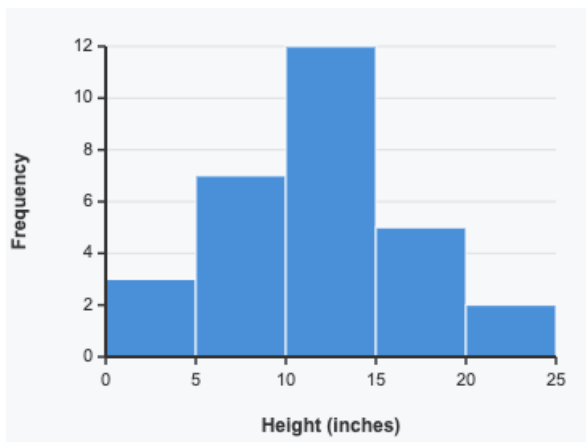
Answer:

Orient students to the frequency table and the empty grid. Point out that the x-axis is already labeled with the intervals from the table and the y-axis counts by 2.

Model drawing the first bar. Ask how many plants are between 0 and 5 inches, then draw a bar over that interval up to a height of 3.

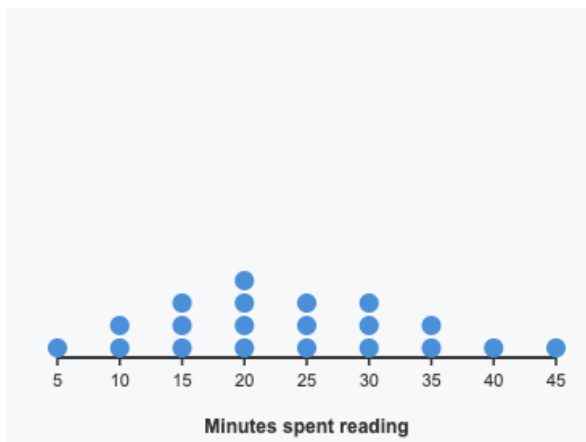
Have students draw the bar for 5 to 10 inches. Ask what height the bar should reach and why.

As students are ready, have them complete the remaining bars on their own. Remind them that the left-end of each interval is included and the right-end is excluded, so a 10-inch plant belongs in the 10 to 15 bin.



3. Compare a histogram and a dot plot of the same data set.

Both displays show the same data set: the number of minutes 20 students spent reading last night.

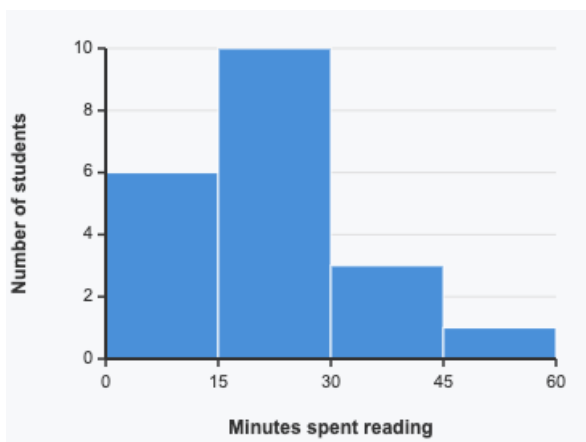


Show students both displays and remind them that each dot on the dot plot represents one student.

For part (a), ask students to find 20 minutes on each display. Point out that the histogram only tells us how many students read between 15 and 30 minutes, not the exact value.

For part (b), ask students which display gives a clearer picture of where most of the data falls. Guide students to see that the histogram groups data into wider bins, which makes the overall shape easier to see at a glance.

For part (c), have students count from each display and check that both give the same answer. Ask students when they might prefer one display over the other. Point out that dot plots show every individual value, while



histograms show the shape of the distribution more clearly when there is a lot of data.

- Which display tells you how many students read for exactly 20 minutes? How many?
- Which display makes it easier to see the overall shape of the data?
- Which display would you use to find how many students read for at least 30 minutes? How many?

Answer: a) The dot plot; 4 students; b) The histogram; c) Either display works. The histogram shows 4 students; the dot plot shows 4 students (3 + 1).

Monitor For

- Do students correctly apply the bin convention (left-end included, right-end excluded) when reading or building bars in Problems 1 and 2?

Instructional Moves

- Before students answer Problem 1 part (a), trace the bars for 30 to 40, 40 to 50, and 50 to 60 pounds with a finger and ask, "Which bars count?"

• In Problem 1 part (b), do students try to give an exact count for 25 pounds, or do they recognize that the histogram only shows a range? • In Problem 2, do students match each bar's height to the frequency listed in the table, including reading the y-axis scale of 2 correctly? • In Problem 3, can students explain that the dot plot shows individual values while the histogram shows the shape of the distribution?	• If a student gives an exact value for Problem 1 part (b), ask, "Can you tell from this bar how many dogs weigh exactly 25 pounds, or only how many weigh between 20 and 30?" • When drawing bars in Problem 2, prompt students to read the y-axis scale aloud: "Each line is 2, so a frequency of 7 falls halfway between 6 and 8." • In Problem 3, ask students to point to the same data value (e.g., 20 minutes) on both displays and describe what each one tells them about it.
Reflection Questions • How did you decide which bars to count for "at least 30 pounds" in Problem 1? <i>Answer: I counted the bars starting at 30 pounds and going right, then added their frequencies.</i> • Why couldn't you tell exactly how many dogs weigh 25 pounds from the histogram? <i>Answer: The histogram groups data into intervals, so the bar for 20 to 30 pounds only tells me the total for the whole range, not any single weight.</i> • When you built the histogram in Problem 2, how did the frequency table help you decide the height of each bar? <i>Answer: The frequency told me how tall each bar should be on the y-axis.</i> • What can you see in a dot plot that you can't see in a histogram, and what can you see in a histogram that's hard to see in a dot plot? <i>Answer: A dot plot shows every individual value. A histogram makes the overall shape of the data easier to see, especially when there are a lot of values.</i>	